FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – I
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**ABC**	19.	**ABC**	37.	**BD**
2.	**BC**	20.	**AD**	38.	**AD**
3.	**BC**	21.	**ACD**	39.	**AD**
4.	**BCD**	22.	**AC**	40.	**ABC**
5.	**BC**	23.	**BC**	41.	**AB**
6.	**BD**	24.	**ABCD**	42.	**BD**
7.	**AB**	25.	**ACD**	43.	**ABD**
8.	**C**	26.	**D**	44.	**C**
9.	**C**	27.	**C**	45.	**D**
10.	**C**	28.	**B**	46.	**D**
11.	**B**	29.	**C**	47.	**A**
12.	**B**	30.	**C**	48.	**D**
13.	**A**	31.	**A**	49.	**B**
14.	**6**	32.	**7**	50.	**3**
15.	**2**	33.	**3**	51.	**6**
16.	**1**	34.	**3**	52.	**6**
17.	**5**	35.	**9**	53.	**5**
18.	**7**	36.	**2**	54.	**7**

Physics

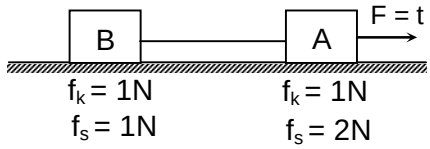
PART – I

SECTION – A

1. ABC

Sol. When t = 3s block just about to move and

acceleration of block given by $a = \frac{t-2}{1}$ $t > 3$



$$\int_0^v dv = \int_3^{10} (t-2)\,dt$$

$$v = \frac{t^2}{2} - 2t\Big|_3^{10} = (50-20) - \left(\frac{9}{2} - 6\right)$$

$$= 30 + 1.5 = 31.5 \text{ m/s}$$

2. BC

Sol. As the tank allowed to fall vertically, so bouyant force becomes zero, so the spring compresses more than its equilibrium compression.

3. BC

Sol. $v^2 = a - bx^2$

differentiating both side w.r.t. time

$2v\frac{dv}{dt} = -2bx\left(\frac{dx}{dt}\right)$ $\frac{dv}{dt} = A$ (acceleration)

$A = -bx$

Hence motion will be S.H.M. $\omega^2 = b$, $4\pi^2 f^2 = b$, $f = \frac{\sqrt{b}}{2\pi}$

$$\text{T.E.} = \frac{1}{2}m\omega^2 (\text{Amplitude})^2 = \frac{1}{2}m\,b\,\frac{a}{b} = \frac{1}{2}ma$$

4. BCD

Sol. Sign between x and t is $\oplus$ hence wave is propagating in negative x direction.

$$\text{wave velocity} = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{3}{1}$$

Max value of y is amplitude, y has maximum value when denominator is least i.e.

when $(x+3t)^2$ is zero

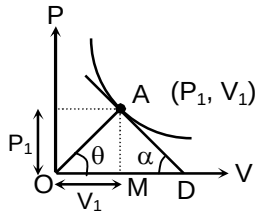
$y = 3m$

shape of pulse is given by $y = \frac{6}{2+x^2}$ $(t=0)$

shape of pulse is symmetrical out y-axis

5. BC

Sol. $\tan\alpha = -$ [slope at (P_1, V_1)] $= \dfrac{P_1}{V_1}$



$\tan\theta = \dfrac{P_1}{V_1}$

$\therefore \ \alpha = \theta$

$OM = MD = V_1$

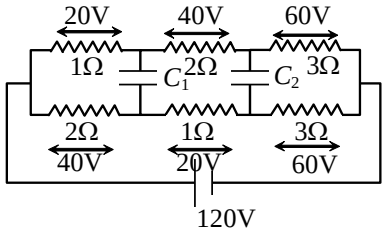
$\therefore$ $OD = 2V_1$

Area of triangle $AOD = \dfrac{1}{2}(AM \times OD) = \dfrac{1}{2}(P_1 \times 2V_1) = P_1V_1 = nRT$

6. BD

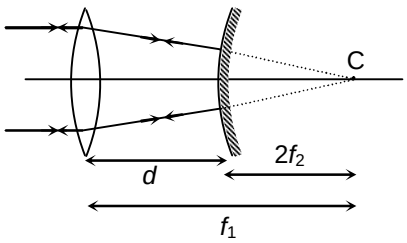
Sol. Now, potential difference across C_1 is 20 V and across C_2 is zero.

$\therefore$ charge stored in C_1 is 40 μC and in C_2 is zero.



7. AB

Sol.



8. C

Sol. Fraction of volume submerged = $d_1 / d_2 = 1$.

9. C

Sol. When ratio of extra weight to total base area is minimum.

10. C

Sol. When ratio of extra weight to total base area is maximum.

11. B

Sol. $Q_{outside}$ = earthed plate will have zero charge on outer surface for its zero potential condition.

12. B

Sol. Earthed plate will have zero charge on outer surface for its zero potential condition.

13. A

Sol. $Q_{outside} = (Q_1+Q_2)/2$, and conservation of charge.

SECTION – C

14. 6

Sol. $\frac{\Delta Q}{\Delta t} = K_A A \frac{dT}{dx}$

$\frac{dT}{dx}$ in conductor A = slope of graph = $\sqrt{3}$

Since both conductors are connected in series, same heat current will flow in A and B.

$K_A (dT/dx)_A = K_B (dT/dx)_B$

$K_B = 6$

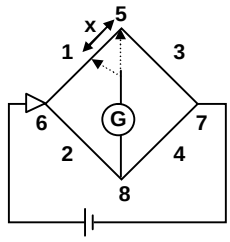
15. 2

Sol. For getting null point

$\frac{R_1}{R_3} = \frac{R_2}{R_4}$

as R2 = R4

R1 = R3



Let the pointer at point 5 is moved to left by distance x to get nul point as shown in the figure. If resistance per unit length of wire 3 is r then that of wire 1 will be 2r.

$(8 - x)2r = x \times 2r + 8 \times r$

$4x = 8$

$x = 2m$

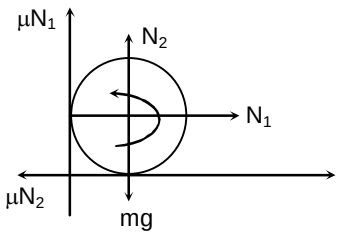
16. 1

Sol. $\omega = \sqrt{\frac{\mu g}{R}} = 1$ rad/sec

17. 5

Sol. $\mu N_1 + N_2 = mg$ . . . (1)

$N_1 = \mu N_2$ . . . (2)



$N_2 = \frac{mg}{(1+\mu^2)}$

$N_1 = \frac{\mu mg}{(1+\mu^2)}$

Torque about centre of mass

$\mu(N_1 + N_2)R = \frac{1}{2}mR^2\alpha$

$\alpha = \frac{2\mu(N_1 + N_2)}{mR} = \frac{2\mu(1+\mu)g}{(1+\mu^2)R}$

$\omega^2 = \omega_0^2 - 2\alpha\theta$

$\theta = \frac{\omega_0^2}{2\alpha}$

$$\theta = \frac{\omega_0^2\left(1+\mu^2\right)R}{2\times 2\mu\left(1+\mu\right)g}$$

Hence number of turns

$$N = \frac{\theta}{2\pi} = \frac{\omega_0^2 R\left(1+\mu^2\right)}{4\mu\left(1+\mu\right)g.2\pi} \quad ; \quad N = \frac{\omega_0^2 R\left(1+\mu^2\right)}{8\mu\pi g\left(1+\mu\right)}$$

18. 7

Sol. Given, voltage V = (100 $\pm$ 5)V

Current I = (10 $\pm$ 0.2)A

According to Ohm's law, V = IR or R = V/I

Taking log of both sides,

Log R = log V – log I

Differentiating, we get

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} - \frac{\Delta I}{I}$$

For maximum error, $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

Multiplying both sides by 100 for taking percentage, we get

$$\frac{\Delta R}{R}\times 100 = \frac{\Delta V}{V}\times 100 + \frac{\Delta I}{I}\times 100$$

Percentage error in resistance R

$$= \frac{\Delta V}{V}\times 100 + \frac{\Delta I}{I}\times 100$$

$$= \frac{5}{100}\times 100 + \frac{0.2}{10}\times 100 = 7\%.$$

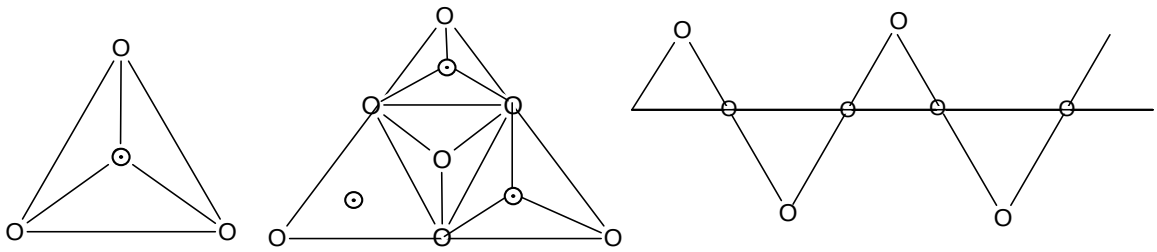
Chemistry

PART – II

SECTION – A

19. ABC

Sol.



20. AD

Sol. $AlCl_3 + 3H_2O \xrightarrow{\text{Hydrolysis}} Al(OH)_3 + 3HCl$

B^{3+} cannot exist in aqueous media

21. ACD

Sol. NO, O_2^{2+} and oxygen having unpaired electrons in molecular orbital.

22. AC

Sol. CN^- and NO^+ total no. of electrons is 14. Hence bond order is 3.

23. BC

Sol. $\underset{(s)}{2C} + \underset{(g)}{3H_2} \rightarrow \underset{(g)}{C_2H_6}$

$\Delta H_f^0 = \left[2 \times \Delta H_{sub}^0 + 3 \times B.E(H-H)\right] - \left[B.E.(C-C) + 6 \times B.E(C-H)\right]$

$\Rightarrow -85 = \left[(2 \times 718) + (3 \times 436)\right] - (x + 6y)$

$\therefore x + 6y = 2829$ 1

Similarly for $C_3H_8(g)$

$2x + 8y = 4002$ 2

Solving (1) & (2), x = 345

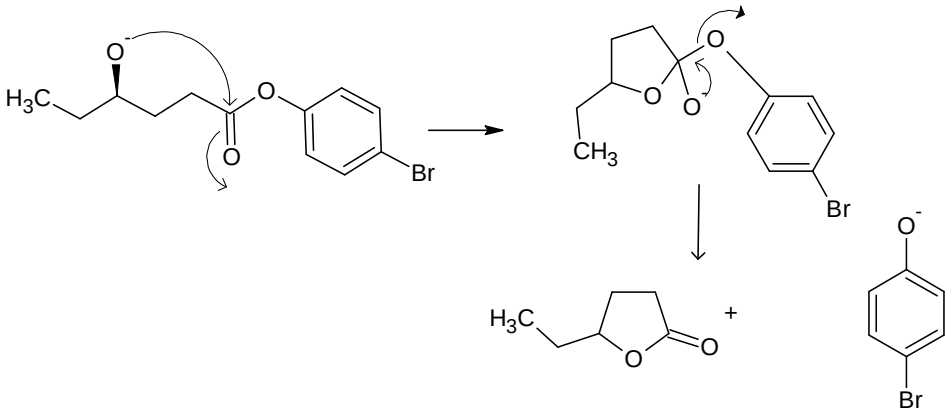
y = 414

24. ABCD

Sol. $\Delta S = \dfrac{\Delta H}{T} = \dfrac{-40,600}{373} = -54.42\,J/K$

25. ACD

Sol.



26. D
Sol. Attack of H^+ on oxygen and then nucleophilic attack of Cl^-

27. C
Sol. Pinawl formation-diracdical formation of Mg complex, then recombination and then hydrolysis.

28. B
Sol. Attack of H^+ on oxygen and then nucleophilic attack of Cl^-

29. C
Sol. An oxalic acid is strong reducing agent having C – C bond length longer than C – C bond length in ethane.

30. C
Sol. $H_3C-\ddot{N}=C=\ddot{O}:$
N is sp^2 hybridized having two bp and one lp and so bent.

31. A
Sol. Silyl isocyanate is linear due to $p\pi$-$d\pi$ bonding in between nitrogen and silicon.

SECTION – C

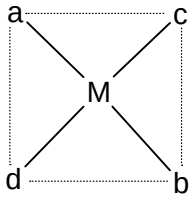
32. 7
Sol. $2KMnO_4 + KI + H_2O \rightarrow 2KOH + 2MnO_2 + KIO_3$
3.5 moles of potassium iodide requires 7 mol of potassium permanganate.

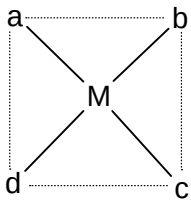
33. 3
Sol. $CoCl_2 + H_2S \rightarrow CoS\downarrow + 2HCl$
$3CoS + 6HCl + 2HNO_3 \rightarrow 3CoCl_2 + 2NO + 3S + 4H_2O$
aqua regia
$KNO_2 + CH_3COOH \rightarrow CH_3COOK + HNO_2$
$CoCl_2 + 2KNO_2 \rightarrow Co(NO_2)_2 + 2KCl$
$Co(NO_2)_2 + 2HNO_2 \rightarrow Co(NO_2)_3 + NO + H_2O$
$Co(NO_2)_3 + 3KNO_2 \rightarrow K_3[Co(NO_2)_6]$
Oxidation number of cobalt in this compound is +3.

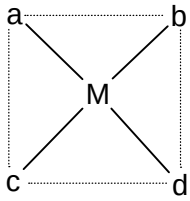
34. 3

Sol. $\left[Pt(NH_3)(NH_2OH)(NO_2)(py)\right]NO_2$

All four ligands are different, so it can be compared to M abcd. Three geometrical isomers are possible as follows







35. 9

Sol. $K_b = 10^{-5}$, $pK_b = 5$

$BOH + HCl \rightarrow BCl + H_2O$

At half neutralization, 50% of the base is converted to its salt, with strong acid HCl, it forms a basic buffer.

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pOH = 5 + \log\frac{[salt]}{[base]}$$

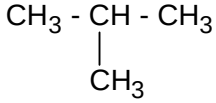
[salt] = [base]

pOH = 5

pH = 14 – pOH = 9

36. 2

Sol.



Mathematics

PART – III

SECTION – A

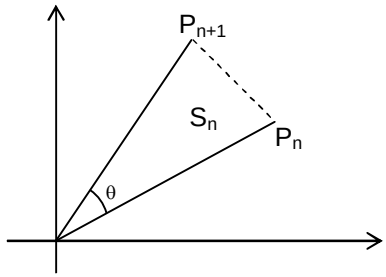
37. BD

Sol. $S_n = \frac{1}{2}|P_n||P_{n+1}|\sin\theta$

$|P_n| = \sqrt{n+1}$

$|P_{n+1}| = \sqrt{n+2}$

$\arg\left(\frac{P_{n+1}}{P_n}\right) = \arg\left(1+\frac{i}{\sqrt{n+1}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{n+1}}\right)$



38. AD

Sol. Let $\vec{a} = \cos\alpha\hat{i} + \sin\alpha\hat{j}, \vec{b} = \cos\beta\hat{i} + \sin\beta\hat{j}, \alpha,\beta,\delta \in [0,2\pi)$

$\vec{c} = \cos\delta\hat{i} + \sin\delta\hat{j}$

$\Rightarrow \cos(\alpha-\delta) = \cos(\beta-\delta) = (\cos(\alpha-\beta))(-\sqrt{3})$

let $A = \alpha-\beta, B = \beta-\delta$

$\Rightarrow \cos A = \cos B = -\sqrt{3}\cos(A-B)$

$\cos A = \cos B \Rightarrow B = 2m\pi \pm A$

$\Rightarrow B = A, B = -A, B = -2\pi + A, B = 2\pi - A$

$\Rightarrow B = A, B = -2\pi + A$ rejected.

$\Rightarrow$ when $B = -A, B = 2\pi - A$

$\Rightarrow \cos A = -\sqrt{3}\cos(2A-2\pi)$ or $\cos A = -\sqrt{3}\cos 2A$

$\Rightarrow \cos A = \frac{1}{\sqrt{3}}, -\frac{\sqrt{3}}{2}$

$\vec{a}.\vec{b} = \cos 2A = \frac{-1}{3}, \frac{1}{2}$

39. AD

Sol. Let $\int_0^t f(u)\,du = F(t) \Rightarrow F'(t) = f(t)$ and $F(0) = 0$

Also $f(t) \geq 0 \Rightarrow \int_0^t f(u)\,du \geq 0 \ \forall\, t \in [0,1]$

$\Rightarrow |f(t)| \leq \sqrt{1+2\int_0^t f(u)\,du}$

$\Rightarrow F'(t) \leq \sqrt{1+2F(t)} \Rightarrow \frac{F'(t)}{\sqrt{1+2F(t)}} \leq 1$ as $f(t) \geq 0$

$\int_0^t \frac{F'(t)\,dt}{\sqrt{1+2F(t)}} \leq \int_0^t dt \ \forall\, t \in [0,1]$

$\Rightarrow \sqrt{1+2F(t)} \le t+1$

$\Rightarrow f(t) \le t+1$ Also $1+2F(t) \le (t+1)^2$

$\Rightarrow 2F(t) \le t^2+2t$

$\Rightarrow F(t) \le \frac{t^2}{2}+t \Rightarrow \int_0^t f(u)du \le \frac{t^2}{2}+t$

40. ABC

Sol. $\int_0^1 x f'(x)dx$, Put $x=\sin\theta \Rightarrow dx=\cos\theta d\theta$

$J=\int_0^{\pi/2} f(\sin\theta)(\cos\theta)d\theta = \int_0^{\pi/2} f(\cos\theta)\sin\theta d\theta$

$2J=\int_0^{\pi/2} f(\sin\theta)\cos\theta + f(\cos\theta)(\sin\theta)d\theta$

$2J \le \int_0^{\pi/2} 1.d\theta \Rightarrow I \le \frac{\pi}{4}$ and equality holds if $f(x)=\sqrt{1-x^2}$

Put $x=y \Rightarrow xf(x) \le \frac{1}{2}$

$\Rightarrow \int_0^1 x f(x)dx \le \int_0^1 \frac{1}{2}dx$

$\Rightarrow f(x)\frac{x^2}{2}\Big|_0^1 - \int_0^1 f'(x)\frac{x^2}{2}dx \le \frac{1}{2}$

$\Rightarrow f(1) - \int_0^1 x^2 f'(x)dx \le 1$

Also $\int_0^1 f(x)dx \le \frac{\pi}{4}$ (Again use By parts)

41. AB

Sol. Using sine law

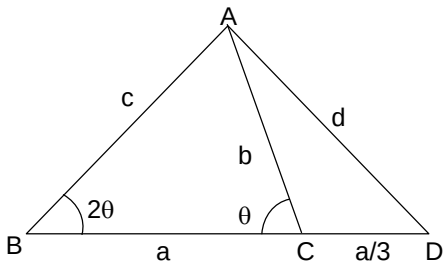
$\frac{b}{\sin 2\theta} = \frac{c}{\sin\theta} = \frac{a}{\sin(\pi-3\theta)}$

$\Rightarrow \frac{b}{2\cos\theta} = c = \frac{a}{3-4\sin^2\theta}$

$\Rightarrow b = 2c\cos\theta$ and $a=(4\cos^2\theta-1)c$

$\Rightarrow ca = b^2-c^2$ (eliminate $\cos\theta$)

Also $\cos(\angle ACB)+\cos(\angle ACD)=0$



$$\Rightarrow \frac{a^2+b^2-c^2}{2ab}+\frac{b^2+\frac{a^2}{9}-d^2}{2\frac{(a)}{3}(b)}=0 \Rightarrow d=\frac{1}{3}(3c+2a)$$

$$\text{Also } \cos\phi=\frac{c^2+d^2-\left(\frac{4a}{3}\right)^2}{2cd}$$

$$\Rightarrow \text{ On manipulation } cd=\frac{a^2}{3\sin^2\frac{\phi}{2}}$$

$$\Delta ABC=\frac{3}{4}(\Delta ABD)=\left(\frac{3}{8}\right)cd\sin\phi \;=\; \frac{1}{4}(BC)^2\frac{\cot\phi}{2}$$

42. BD

Sol. Given $f(j)-j<f(i)-i\ \forall i<j$

Let $g(k)=f(k)-k \Rightarrow g(k)$ is decreasing function. Also

$1\le f(k)\le 2018$

$.\Rightarrow 1-k\le g(k)\le 2018-k$.

Also $g(k+1)$ will always be in the desired range as long as g (k) is in the desired Range.

We are supposed to select $g(1),g(2),.....g(2018)$ values in decreasing order from $[-2017, 2017]$

$\Rightarrow {}^{4035}C_{2018}$ ways.

$${}^{2n}C_n=\sum_{i=0}^{n}\left({}^nC_i\right)^2 \;\Rightarrow {}^{4035}C_{2018}=\frac{4035}{2018}\;{}^{4034}C_{2017}$$

43. ABD

Sol. Let $h(x)=f(f(f(x)))=x\ \forall\ x\in R$

$\Rightarrow h(x)$ is a strictly increasing and bijective

Also if f is strictly decreasing, $f(f)$ is increasing function,

$\Rightarrow f(f(f(x)))$ is decreasing, hence contradiction.

Also if for $x_2>x_1$, $f(x_2)=f(x_1)$

$\Rightarrow f(f(x_2))=f((f(x_1)))$

$\Rightarrow f(f(x_2))-f(f(f(x_2)))$

$\Rightarrow h(x_2)=h(x_2)$ not possible for $x_2\ne x_1$

If f is a strictly increasing function

$\Rightarrow f(f)$ is a strictly increasing function

$\Rightarrow f(f(f))$ is a strictly increasing function

Also Range of $h(x)=R\subseteq$ Range of $f(x)$

$\Rightarrow f(x)$ is on to $\Rightarrow$ hence bijective.

Also $h(0)=0 \Rightarrow f(f(f))=0$ and $f(x)$ and $f(f(x))$ are inverse of each other $\Rightarrow f(0)=0$ and increasing i.e. Non – zero slope. Hence will cut $y = x^2$ at two distinct points.

44. C
45. D
46. D

(Sol. 44 to 46)

$A + C = B + D \Rightarrow D : (2,1,8)$

$\overline{BA} = (1,-1,5), \overline{BC} = (0,2,4)$

Equation of plane π is $7x + 2y - z - 8 = 0$

Equation of line PF $\bar{r} = (2,1,0) + \delta(7,2,-1)$

$\Rightarrow \delta = (2+7\delta, 1+2\delta, -\delta)$, for other point put $\delta = -1$

For point F: $7(2+7\delta) + 2(1+2\delta) + \delta - 8 = 0$ i.e. satisfying in Plane π

$$\Rightarrow \delta = \frac{-4}{27}$$

$$F : \left(\frac{26}{27}, \frac{19}{27}, \frac{4}{27}\right)$$

$$I : 2F - P : \left(\frac{-2}{27}, \frac{11}{27}, \frac{8}{27}\right)$$

Foot of perpendicular from A to BC:

Equation of BC : (1, 0, –1) + λ (0, 1, 2)

$: (1, \lambda, -1+2\lambda)$

For foot of perpendicular $\left(\hat{j} + 2\hat{k}\right).\left(\hat{i} + (-1-\lambda)\hat{j} + (5-2\lambda)\hat{k}\right) = 0$

$$\lambda = \frac{9}{5} \Rightarrow \left(1, \frac{9}{5}, \frac{13}{5}\right)$$

Line L intersects plane π at (1, 0, –1) (Solve plane and line L)

Equation of Projection of line L is passing through F and (1, 0, –1)

Equation of mirror image of line L in Plane is line passing through I and (1, 0, –1)

Let ϕ be the acute angle between line L and plane π

$$\Rightarrow \cos\theta = \left|\frac{\left(\hat{i} + \hat{j} + \hat{k}\right).\left(7\hat{i} + 2\hat{j} - \hat{k}\right)}{\sqrt{3}.\sqrt{54}}\right| \text{ where, } \theta + \phi = \frac{\pi}{2}$$

$$= \left|\frac{8}{9\sqrt{2}}\right|$$

$$\Rightarrow \phi = \sin^{-1}\left(\frac{8}{9\sqrt{2}}\right)$$

For distance between AB and CD say H

Area of parallelogram ABCD $= \left|\vec{AB} \times \vec{BC}\right| = 2\sqrt{54}$

$(H)(BA) = 2\sqrt{54}$ when $BA = \sqrt{27}$

47. A
48. D
49. B

(Sol. 47 to 49)

Equation of tangent is $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$

a = 4, b = 3

$\Rightarrow A:(4\sec\theta,0), B:(0,3\operatorname{cosec}\theta)$

$\Delta OAB=\frac{1}{2}(45\sec\theta)(3\operatorname{cosec}\theta)=\frac{12}{\sin 2\theta}$

$(\Delta OAB)_{min}=12$ where $\theta=\frac{\pi}{4}$

$P:\left(\frac{4}{\sqrt{2}},\frac{3}{\sqrt{2}}\right), A:(4\sqrt{2},0), B:(0,3\sqrt{2})$

Equation of tangents: $\frac{x\left(\frac{1}{\sqrt{2}}\right)}{4}+\frac{y\left(\frac{1}{\sqrt{2}}\right)}{3}=1$

$\Rightarrow 3x+4y=12\sqrt{2}$

Let $Q:(\alpha,\beta)\Rightarrow 3\alpha+4\beta=12\sqrt{2}$

Equation of CD : $x\alpha+y\beta=12\Rightarrow x\left(\frac{12\sqrt{2}-4\beta}{3}\right)+y\beta=12$

$\Rightarrow (12\sqrt{2}x-36)+\beta(3y-4x)=0$

$\Rightarrow$ Point of concurrency: $\left(\frac{3}{\sqrt{2}},\frac{4}{\sqrt{2}}\right)$

Also distance of (0, 0) from $3x+4y-12\sqrt{2}=0$ is $\frac{12\sqrt{2}}{5}$ as radius of circle is $\sqrt{12}$

$\Rightarrow \frac{\text{distance}}{\text{radius}}=\frac{\sqrt{24}}{\sqrt{25}}<1$

$\Rightarrow$ cuts at 2 points.

Let (h, k) be the circumcentre of ΔQCD

$\Rightarrow (h,k)$ is midpoint of (0, 0) and (α,β)

$\Rightarrow 2h=\alpha,\ 2k=\beta$

Also $3\alpha+4\beta=12\sqrt{2}\Rightarrow 3h+4k=6\sqrt{2}$

For R equation of director circle of ellipse
For (iv) equation of director circle of circle
For (S) Any line passing through one focus of ellipse will pass through other focus after reflection.

SECTION – C

50. 3

Sol. $y = mx$ touches circle

$\Rightarrow D = 0$ for $(m^2+1)x^2 - 2a(m+2)x + 4a^2 = 0$

$\Rightarrow 4a^2(m+2)^2 = 4(m^2+1)(4a^2)$

$\Rightarrow m = 0, \frac{4}{3}$

Let $A:(x_1, m_1x_1), B:(x_2, m_2x_2)$

$AB^2 = 9 \Rightarrow (x_1 - x_2)^2 + (m_1x_1 - m_2x_2)^2 = 9$ where

$x_1 = \frac{(m_1+2)}{m_1^2+1}a,\ x_2 = \frac{(m_2+2)}{m_2^2+1}a,\ m_1 = 0,\ m_2 = \frac{4}{3}$

get $a = \frac{3\sqrt{5}}{4}$

51. 6

Sol. Let $1-x = a,\ 2-y = b,\ 3-z = c \Rightarrow a,b,c > 0$

$E = (abc)\left(3 - a - \frac{b}{2} - \frac{c}{3}\right)$

$E = 6(a)\left(\frac{b}{2}\right)\left(\frac{c}{3}\right)\left(3 - a - \frac{b}{2} - \frac{c}{3}\right)$

Apply $AM \geq GM$

$\Rightarrow \frac{a + \frac{b}{2} + \frac{c}{3} + \left(3 - a - \frac{b}{2} - \frac{c}{3}\right)}{4} \geq \left[a\left(\frac{b}{2}\right)\left(\frac{c}{3}\right)\left(3 - a - \frac{b}{2} - \frac{c}{3}\right)\right]^{1/4}$

$\Rightarrow E \leq \frac{243}{128}$ and equality holds for $a = \frac{3}{4},\ b = \frac{3}{2},\ c = \frac{9}{4}$

$\Rightarrow (x_0, y_0, z_0) \equiv \left(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right)$

52. 6

Sol. $\sin x + \sin y = \sin(x+y)$

$\Rightarrow$ Either $\sin\left(\frac{x+y}{2}\right) = 0$ or $\cos\left(\frac{x+y}{2}\right) = \cos\left(\frac{x-y}{2}\right)$

$\Rightarrow x + y = 2m\pi$ or $\frac{x}{2} = p\pi$ or $\frac{y}{2} = k\pi\ (m,p,k \in \text{Integer})$

$\Rightarrow x + y = 2m\pi$ or $x = 2p\pi$ or $y = 2k\pi$

Plot and get (0, 1), (0, –1) , (–1, 0), (1, 0) $\left(\frac{1}{2}, \frac{-1}{2}\right), \left(\frac{-1}{2}, \frac{1}{2}\right)$

53. 5

Sol. Equation of normal at point $T\left(t,\frac{t^2}{2}\right)$ to C is $y-\frac{t^2}{2}=\frac{-1}{t}(x-t)\Rightarrow y=\frac{-x}{t}+\frac{t^2}{2}+1$

If circle S touches curve C at T.

$\Rightarrow (x(t),y(t))$ must lie on normal

$\Rightarrow y(t)=\frac{-x(t)}{t}+\frac{t^2}{2}+1$

Also distance of centre of circle from T equals radius = distance of centre of circle from x – axis.

$\Rightarrow (x(t)-t)^2+\left(y(t)-\frac{t^2}{2}\right)^2=(y(t))^2$

$x(t)=\frac{t\left(1+\sqrt{t^2+1}\right)}{2},\ y(t)=\frac{t^2+1-\sqrt{t^2+1}}{2}$

$\Rightarrow x(t)y(t)=\frac{1}{4}t^3\sqrt{t^2+1}$

$\lim\limits_{r\to 0^+}\int\limits_r^1\frac{1}{4}t^3\sqrt{t^2+1}\,dt$

Let $\sqrt{1+t^2}=z\Rightarrow 2tdt=2z\,dz$

$\Rightarrow \lim\limits_{r\to 0^+}\int\limits_{\sqrt{1+r^2}}^{\sqrt{2}}\frac{1}{4}(z^2-1)z^2dz=\frac{\sqrt{2}+1}{30}$

54. 7

Sol. Let E_1 : first ball drawn red from Urn A and 2nd ball drawn black from B.

E_2 : first ball drawn while from A and 2nd ball drawn black from B

E_3 : first ball drawn red from B and 2nd ball drawn black from B

E_4 : first ball drawn black from B and 2nd ball drawn black from B.

$P=\frac{P(E_1)+P(E_3)}{P(E_1)+P(E_2)+P(E_3)+P(E_4)}$

$P(E_1)=\left(\frac{1}{2}\right)\left(\frac{2}{6}\right)\left(\frac{3}{6}\right)\left(\frac{1}{2}\right)=\frac{6}{144}$

$P(E_2)=\left(\frac{1}{2}\right)\left(\frac{4}{6}\right)\left(\frac{1}{2}\right)\left(\frac{3}{6}\right)=\frac{12}{144}$

$P(E_3)=\left(\frac{1}{2}\right)\left(\frac{3}{6}\right)\left(\frac{1}{2}\right)\left(\frac{3}{5}\right)=\frac{9}{120}$

$P(E_4)=\left(\frac{1}{2}\right)\left(\frac{3}{6}\right)\left(\frac{1}{2}\right)\left(\frac{2}{5}\right)=\frac{6}{120}\ \Rightarrow P=\frac{7}{15}$